

Forey Forest Intelligence Challenge

What is Forey?

Forey develops a drone-based forest inventory and analytics service built on geospatial data. Forest areas are surveyed using LiDAR, standard cameras or multispectral cameras. The collected data is then processed into forest and terrain information that can be used on maps and in 3D environments.

Forey aims to produce forest information that is as accurate and up to date as possible, down to individual trees. The data can support forest management and harvesting plans, forest property valuations, timber sales and the detection of changes in forests.

Forey's current workflow can be summarised as follows:

A work area is defined

- a drone flight is planned for the area
- an operator collects LiDAR and image data
- the data is transferred to the cloud
- forest and terrain information is generated from the data
- the results are presented on a map, as a 3D model and as key metrics

In this hackathon, your task is to take the same technology into new applications.

Challenge

Build a concept and a working demo that turns data collected from forests into a solution that improves safety, preparedness, infrastructure resilience or operational decision-making, with the potential to become a new business.

The solution may use, for example:

- drone imagery
- LiDAR point clouds
- multispectral imagery
- satellite imagery
- individual-tree data
- forest resource and vegetation data
- terrain and elevation models
- weather, drought and wind data
- road, water and infrastructure data
- other open or synthetic data

Each team selects one of the main categories described below. Categories may be combined, but the solution must have one clearly defined main problem, user and decision that it improves.

The examples under each category are provided as inspiration. They are not ready-made task lists and do not exclude other ideas.

1. Wildfires and environmental risks

How can forest and terrain data be used to prevent or prepare for wildfires and other environmental risks, and to assess their impacts?

2. Protecting critical infrastructure

How can data on individual trees, forest structure and changing conditions be combined to identify infrastructure risks and plan preventive action?

3. Disruptions and resource coordination

How can forest data support the right decisions during the first hours after a storm, wildfire or other disruption?

4. Terrain accessibility and safe routing

How can forest, vegetation and terrain data be used to create safe and feasible routes for different users and vehicle classes?

5. Preparedness, logistics and defensive security applications

How can data collected from forests support societal preparedness, logistics, rescue operations, training or other defensive activities?

Deliverables

What should the team build?

1. A concept and a clearly defined user
2. A working technical demo
3. A description of the data and analysis
4. The technical architecture and scalability approach
5. The commercial opportunity
6. A final presentation

The presentation must answer the following questions:

1. What problem is being solved?
2. Who uses the solution?
3. What data does the solution use?
4. What did the team build?
5. How does the analysis or decision-making logic work?
6. What decision does the solution improve?
7. What is Forey's role?
8. Who would pay for the solution?
9. What would the first pilot look like?